

Bio+MedVis Challenge @ IEEE VIS

Made of Stars

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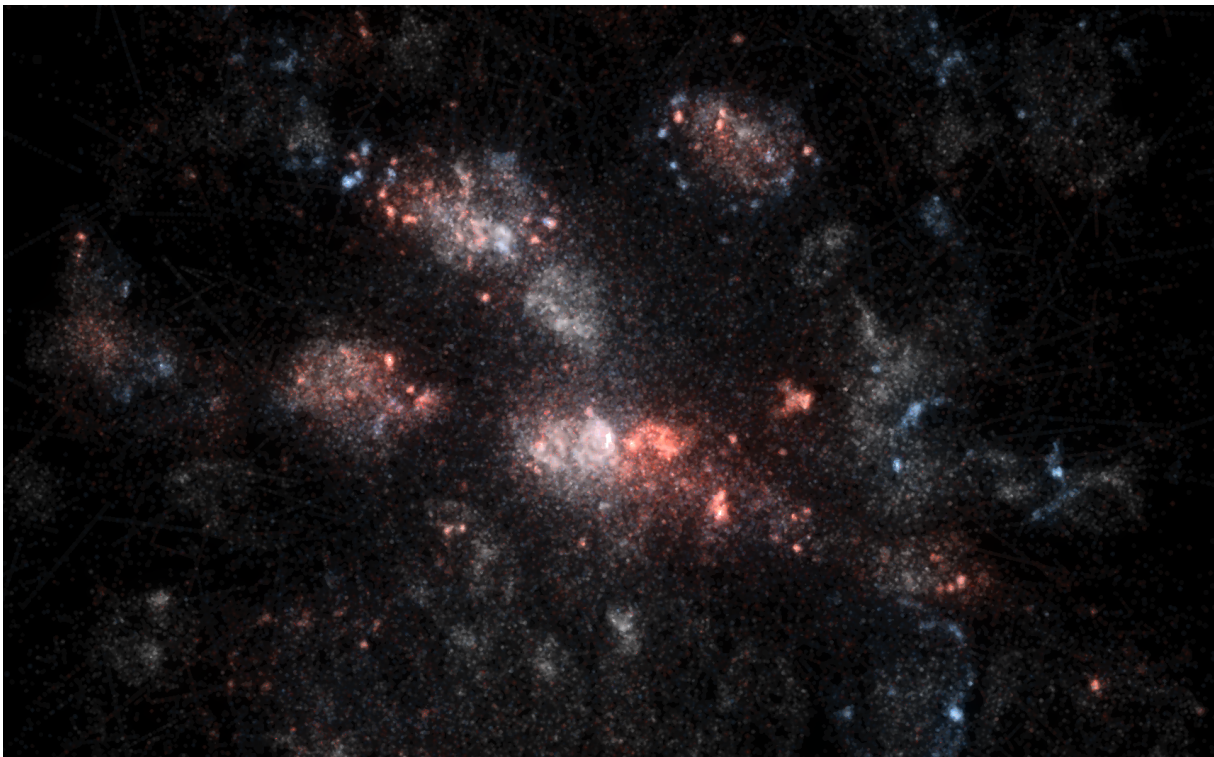


Figure 1: Made of Stars with 3 microscopy channels (Catalase, Hoechst, CD4). Image obtained after averaging 30 individual frames.

ABSTRACT

We introduce *Made of Stars*, a dynamic volume visualization that fuses a flocking algorithm, synchronized firing, and gradient-derived force fields to produce a dynamic and engaging, biologically meaningful depiction of confocal microscopy data. The particle motion is derived from a statistical inference system based on static multi-channel microscopy volumes. We evaluate the effectiveness of the visualization using eye-tracking. An audio channel is introduced to support the perception of global image features. Our application supports the visualization of cell membranes and the uptake of marker proteins to reveal insights into cell morphologies and cell interactions.

Index Terms: Computing methodologies—Computer graphics—Image manipulation—Image-based rendering; Mathematics of computing—Probability and statistics—Probabilistic inference problems

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1 MOTIVATION

As part of the Bio+MedVis microscopy challenge at IEEE VIS 2025 our contribution provides a novel interactive and dynamic visualization of a multi-channel confocal microscopy [5] dataset.

Volume visualization techniques map a static volumetric dataset to pixel intensities using color and transparency. The user is left to visually explore a static volume visualization [8]. We propose that such static displays can limit user engagement specifically in complex scenes where the goal is data exploration. Continued interaction with the scene might be required to extract additional information. Dynamic volume renderings, can be more engaging as motion captures the viewer’s attention [3] and highlights individual features of the data [7].

Here we explore the use of a dynamic volume rendering technique for static volumetric data. Can such a dynamic rendering capture user attention for longer than a static rendering of the same content? In order to test this hypothesis we extend volume visualization with animations using a flock of birds [6] algorithm to generate visually engaging scenes (see figure 2). To our knowledge this technique is a new indirect volume rendering approach. Other approaches for indirect volume rendering convert volume data to, for example, an intermediate representation such as surface meshes. Our approach converts the volume data into a physical statistical inference system that the flock of birds algorithm samples over time. We animate the sampling process to create a dynamic volume visualization for static

data.

2 METHODS

We provide details of the visualization process, complimentary auditory information, and a validation using eye-tracking.

Visualization Procedure We introduce a stellar particles (“spxel” / spek-sl) - based dynamic visualization for multi-channel confocal microscopy data. The particles interact with each other using the flocking algorithm [1] and sample the microscope image intensity using a gradient derived force field. By moving along the image gradient spxels collect in regions of higher image intensity that represent immunofluorescence labels at a subcellular level. To enhance visual grouping in 3D, spxels exhibit synchronized firing based on distance. We present a web-application implementing spxel-based dynamic volume rendering. It adopts an existing flocking and synchronized firing software [1, 4] and extends it with an statistical inference system and data sonification.

To make the flocking algorithm sensitive to the volumetric data, we add a force term based on the gradient of each microscopy channel computed using the finite difference method after spatial smoothing. Particles are color coded and assigned to individual microscopy channels. Over time particles individually explore the data by interacting with each other and by moving into regions of positive gradient (from dark to bright image intensity). The application displays this dynamic in real-time and allows for user interactions. To support visual grouping in a crowded environment, animated particles exhibit synchronized firing based on spatial proximity.

Auditory Application The information content of multi-channel volumetric image visualization can be high. To further support inclusiveness and perception of overall properties of the data like compactness and channel-cross talk we add a data sonification module. The audio channel uses pentatonic tones linked to particle properties speed, position and firing frequency.

Validation We use eye tracking to evaluate the effectiveness of the visualization (GazePoint-3 using PsychoPy v2025.1.1) with one pilot participant. PsychoPy programs and scripts are available on OSF [<https://osf.io/gsjpr/>]. We compared synchronized and desynchronized firing to establish visual preference, and displays with particle motion influenced by flocking plus microscopy data and by flocking only to establish behavioral differences in gaze fixation. Videos were created with the prototype application (3000 stellar particles) at a resolution of 600x600 pixels for 1 min. The participant was instructed to passively view the video. Each of the conditions lasted for 30 seconds. The participant viewed the stimuli on a 1920x1200 screen seated 50 cm away. Fixations were recorded at an interval of 16.4 ms. Standard eye tracking calibration used the 13 calibration points procedure. Figure 6 illustrates the results of the eye tracking validation.

3 DISCUSSION

Our flocking algorithm can create coherent motion on multiple scales. Attention can either be captured by individual spxel, by regions with tens to hundreds of spxels or, by large scale patterns that are created predominantly by the synchronized firing. Demonstrating successful engagement, viewers have anecdotally reported random or ambiguous stimuli (pareidolia) early on in the simulation with grid- and string-like patterns. Due to the number of possible algorithmic settings more work is needed to explore interactions between synchronized firing, flocking and gradient generated force fields. This can optimize the visual appearance, retention and how faithful the volume data is represented.

Our data sonification links tone quality and notes played to the data by assigning higher frequencies to spxels further away from the bounding box center. Data clustering and average speed of spxels are additional channel-based properties that influence both the perceived pitch and timbre of different notes.

Following [2] we create an image-based energy function and a physical statistical inference system that determines a Gibbs distribution converging close to the image distribution. Different from Gibbs sampling, the energy function is sampled using the flocking algorithm with groups of locally interacting spxels exhibiting chaotic behavior. As energy dissipates the simulated particles come closer to the stationary data distribution. The velocity of particles represents the inverse of the likelihood function, of how well our visualization matches the data. The prior distribution is analogous to the image intensities and our posterior distribution is the final position of particles following the present dynamic algorithm.

Due to the initial conditions of all spxels being placed on a sphere the final channel-dependent configuration is only obtained after a long relaxation time of the system. Switching the system from one channel to another can take minutes until a stable state is reached again. A better initial placement using methods such as adaptive rejection sampling or Metropolis-Hastings could provide starting positions closer to the final distribution to reduce channel switching time. A limitation is that a pre-processing step is required to first extract high-resolution volumetric images from the pyramidal tiff and convert them into the compressed binary gradient format used by the application. A backend able to perform this step in real-time would enable the user to move outside of the currently fixed field of view.

The approach allows for customizations based on a large variety of image properties. Potential applications range from improving detection for clinical applications to improvements in attention capture for the viewer with auditory stimulation. A head-to-head comparison with existing well established and known techniques has been intentionally postponed to allow the approach to mature.

Made of Stars is available as source code and can be experienced live at the following link <https://github.com/HaukeBartsch/MadeOfStars/>.

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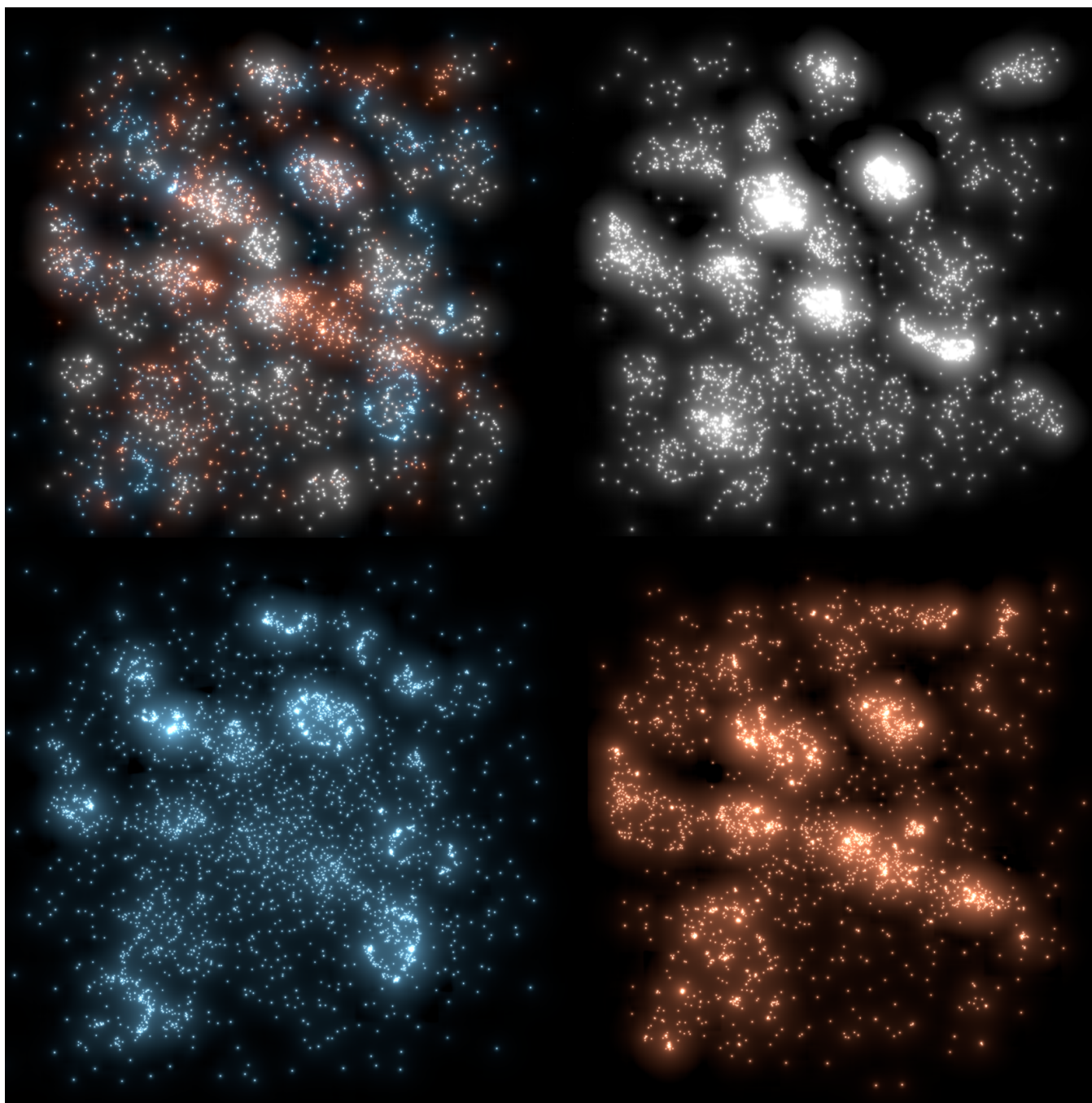


Figure 2: Multi-channel visualization (top left) of dataset LSP13626 melanoma-in-situ and individual channels (top right, Hoechst; bottom left, CD4 and bottom right, Catalase) from the same region. Number of spxels 3,000 in all 4 views.

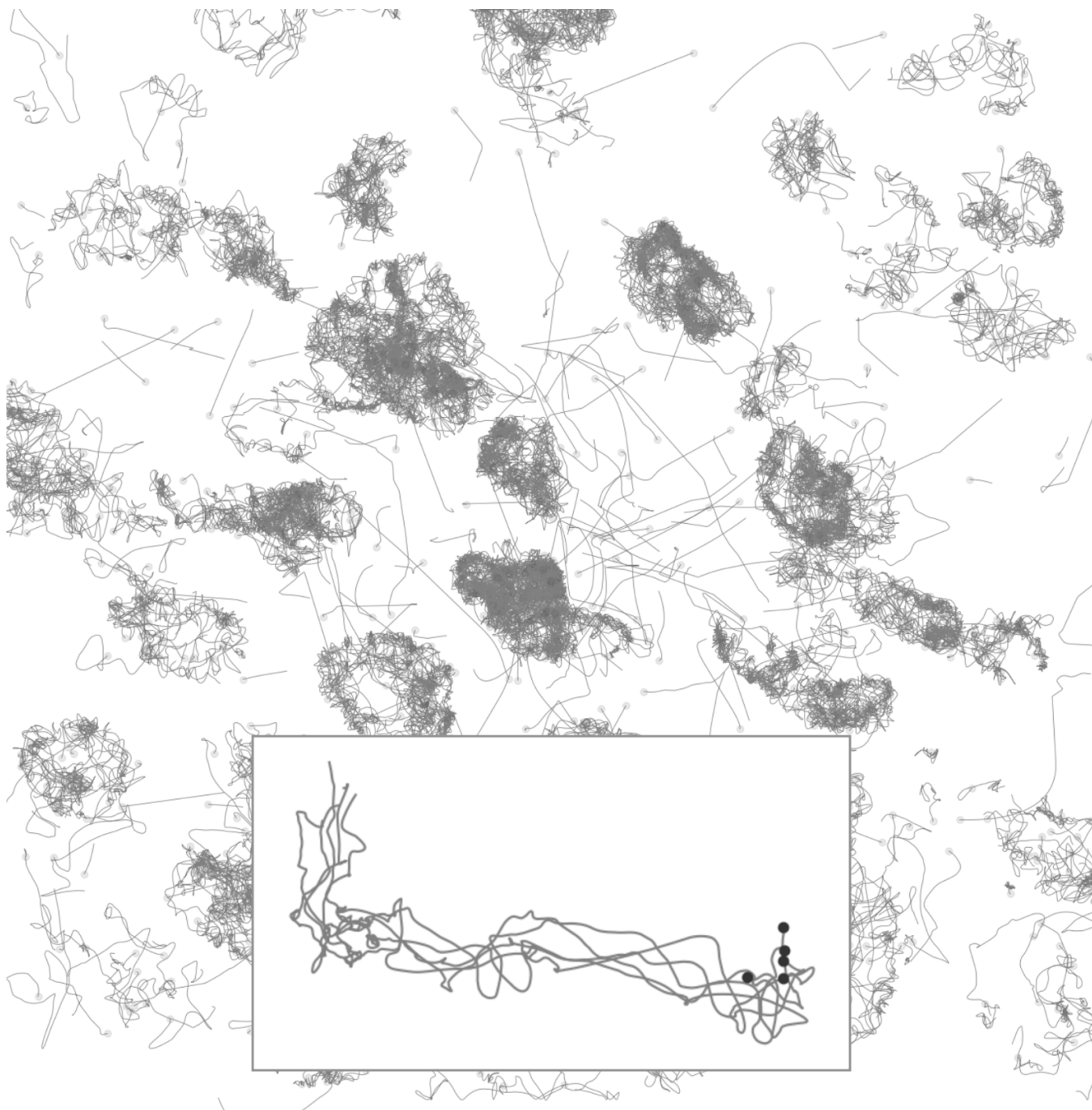


Figure 3: Time evolution of stellar particles (spxels) tracing a confocal imaging stack stained with Hoechst nucleic acid, a cell-permeant nuclear counterstain. A gray circle is added to the last sampled time point. Spxels are either tightly bound to stained structures or diffuse freely in the volume. The inset shows a separate simulation of five particles interacting locally as a quasi-particle (time advances from top left to bottom right). The flocking algorithm keeps them bound to each other as they explore the underlying volume.



Figure 4: The g pentatonic scale. The signal spectra are mapped to the g pentatonic scale across 4 octaves (pictured are two). The pentatonic scale was chosen for its lack of half step intervals and thereby pleasant, dissonance-free sound no matter the permutations of notes (chords) played. The root frequency of 196hz (G3) was chosen as a good baseline providing deep notes and a comfortable range for higher signals to be heard.

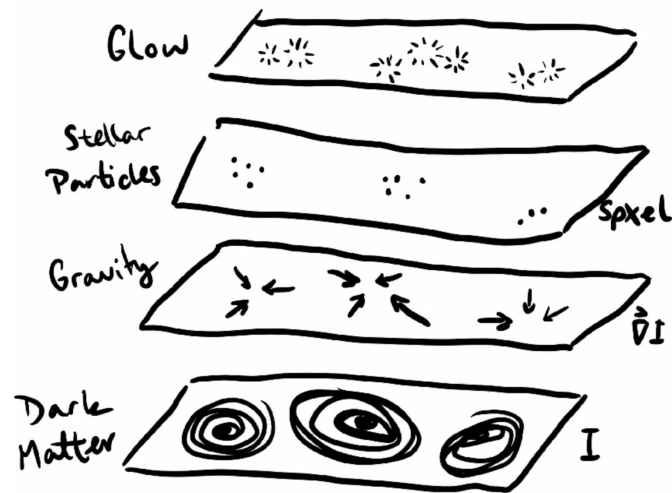


Figure 5: Our initial design of this visualization. We treat image data I as a form of dark matter that affects stellar particles via a gravitational force field ($\vec{\nabla} I$). Our particles move based on particle - particle interactions and they move also based on the gravitational force. Each particle is visually grouped with its neighbors based on distance and a firing (glow) circle.

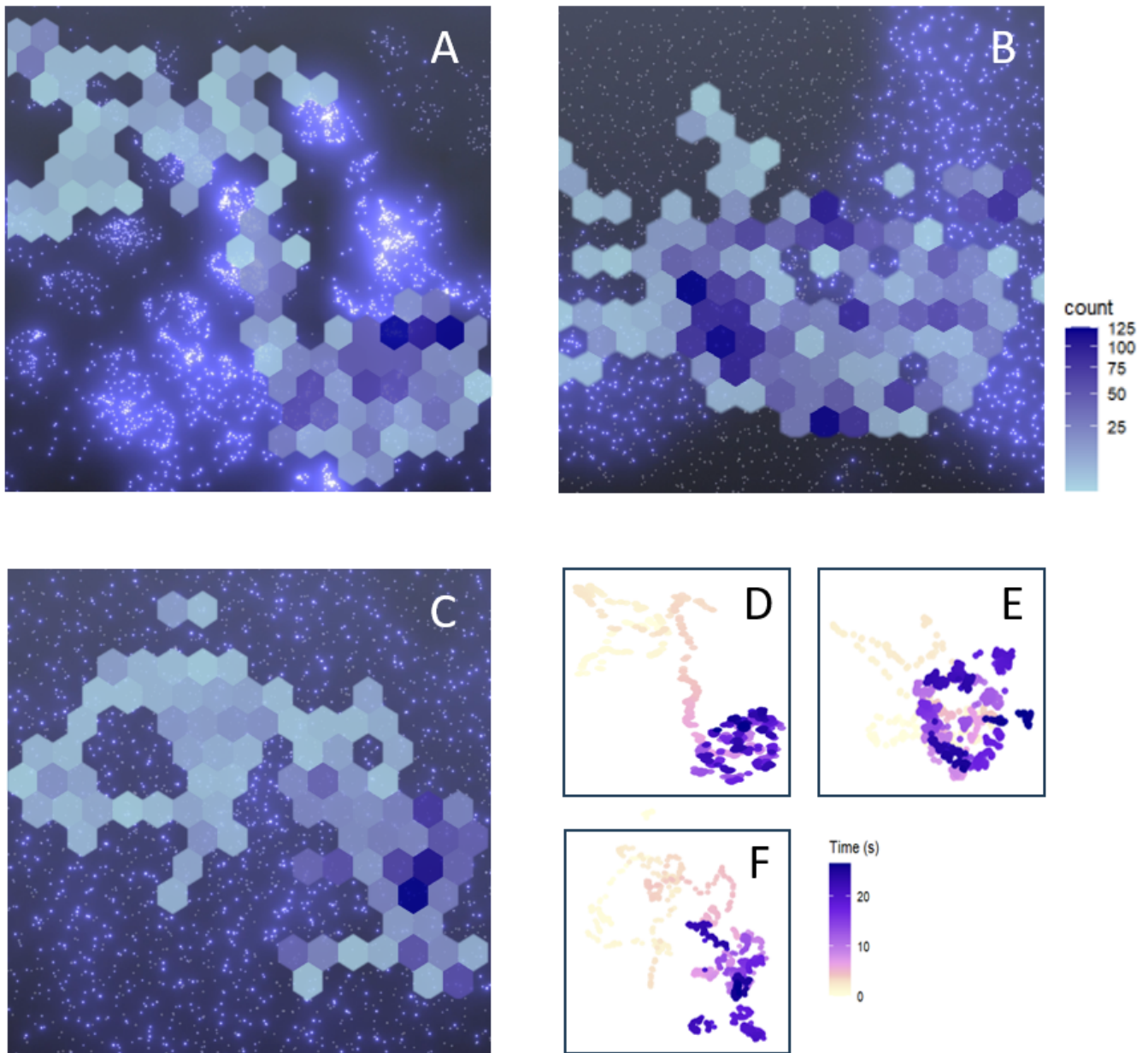


Figure 6: Three eye-tracking conditions shown using hexbin density maps superimposed on still images of the video conditions using one channel. The density of fixations during the data collection time is depicted as the hue of the hexagons, where darker hexagons represent a higher density of fixations. An absence of hexagons represents a lack of data (i.e., the participant did not look at these parts of the screen during the 30 second target time). Condition A: 30 seconds synchronized, with data, Condition B: 3 seconds looped and synchronized without data, Condition C: 30 seconds desynchronized without data, and D, E, F: maps of fixations during the respective condition. The gaze fixation locations greater than 10 ms shown across time (color). Prior to each 30 second video clip condition, the participant underwent a calibration of the eye tracking and a validation procedure. The standardized 13-point eye tracking calibration and validation procedures in the GazePoint plug-in for the PsychoPy software were applied. The participant was seated 50 cm away from the 1290x1200 screen and instructed to fixate on each calibration point (black circles with a red center of 1-degree diameter) presented on a grey background for 500 ms with a 250 ms inter-stimulus interval during which the target moved to a new location semi-randomly until all 13 targets were filled. An identical validation procedure immediately followed this calibration procedure. No drift checks were performed during data recording. By comparing gaze fixations in reaction to single channel visualizations that differ in both synchronization and whether or not they are based on real or simulated data. Over the course of 30 seconds, the participant explores around the display in all conditions. Of note in the experimental condition A is that the participant has followed a similar path from the bottom right to the top left more than once, as can be seen by the color gradients.